

1.	Display line 10 to 15 from file X1 – head –n 15 x1 tail –n 5
2.	Display last word of the file X1 – tr ‘ ’ ‘\012’ < x1 tail –n 1
3.	Count the length of last but one line of x1. – tail –n 2 x1 head –n 1 wc –c
4.	Count the number of words in line 4 to 6 of file f1.—head –n 6 f1 tail –n 3 wc -w
5.	Delete all special characters from the file f1. Tr -d ‘ ?/\$%*&’ ‘ ‘<f1 or Tr –cd ‘[a-z][A-Z][0-9]’<x
6.	Count total number of spaces in a file F1. Tr –cd ‘ ‘ <x wc –c
7.	Change the delimiter of file stud.lst from to ;tr ‘ ’ ‘;’ <x
8.	Display line number before each line. Cat –n x
9.	To remove a directory tree dir1/dir2/dir3 using single command. Rmdir -r dir1 Rmdir –p dir1/dir2/dir3
10.	Move all files begin with digits from parent directory to current directory.- mv [0-9]* a2/
11.	Display all uniq word of file F1.sort –u f1
12.	Double space the file X1.cat x1 pr –d –t
13.	Count the occurrence of each word. Sort f1 uniq –c
14.	Display last accessed file. Ls –lt head –n 1
15.	Display last three modified files.ls –lut head –n 3
15.	Display smallest file of file X1.ls –S tail –n 1
16.	Write a command to remove the repeated lines from the file F1. Sort f1 uniq –u
17.	Display file size and file name of all files in current directory.ls –l tr –s ‘ ’ cut –d ‘ ’ –f 5,9
18.	Display all available users and their home directory.cat /etc/passwd cut –d ‘:’-f1,6
19.	Count number of files of current directory start with alphabets and end with digits. Ls [a-zA-Z]*[0-9]

20	Count number of login users. Who wc -l
21	Count number of vowels in a file X1. Cat b tr -cd 'aeiou' wc -c
22	Display all lines of file F1(do not use pr and cat) tail -n +0 f1
23	List out all files of current directory(do not use ls) echo *
24	Write a commands equivalent to wc. Cat -n x tail -n 1 cut -f 1 Tr ' ' '\012' < x cat -n tail -n 1 cut -f 1
25	Reverse the file F1. nl f1 sort -nr cut -f 2
26	Display not logged in users. Who ect/passwd/comm. Who cut -d ' ' -f 1 sort >c1 Cat /etc/passwd cut -d ':' -f 1 sort >c2 Comm -13 c1 c2
27	Count the number of differ characters in file F1 and F2.cmp -l f1f2 wc
28	Display the length of the longest line. Wc -L stud
29	Remove all the files except .log rm *.(!l)(!o)(!g)
30	Create a directory structure d1/d2/d3 Mkdir -p d1/d2/d3
31	Create a directory Name TYBCAsem5 having initial permission to owner group and other is read and execute. Mkdir -m a=rw tybcasem5
32	Copy f1 to f2 without any warning Cp -f f1 f2
33	Display name of file that begin with dot(.) Ls -ld .*
34	Copy all files that do not end with digit to directory d1 present under the parent directory Cp *[^0-9] d1
35	Copy only those files to directory d1 which begin with 'f' Cp f* d1
36	Copy files of the current directory to directory d1 in which file contains digit any where in the file. Cp *[0-9]* d1
37	Copy 5-character file name of current directory to directory d1 present under the working directory Cp ????? d1
38	Copy 5-character file name that start with alphabet and ends with digit to directory d1 present under the working directory Cp [a-zA-Z]???[0-9] d1

39	Remove all files of current directory whose name start with ! Cd d1 rm -f \!*
40	Remove all files from d1 which do not contain first character as alphabet Cd d1 rm -f [^a-zA-Z]*
41	Remove all files from d1 which do not contain first character is * Cd d1 rm -f **
42	Swith to directory two level up Cd ../..
43	Display all hidden file of working directory without LS command Echo \.*
44	Rename f1 to f2 and move it to d1 directory. mv f1 f2

1. Cat v1>v2
2. Cut -d ":" -f 1 /etc/passwd|grep [0-9]\$
3. tr -s "" "\012" <p| cat -n|tail -1|cut -f 1
4. ls -1|head -1
5. tr -c '[0-9a-zA-Z]' '*'<stud
6. tr 'abcde' 'xy' <stud
7. **tr " '\012'<x|sort|uniq -c|sort -n|tail -1**
8. **printf%x,192**

-
1. grep '^..\.*\.\$' p
 2. grep -E '^+\$\$' p
 3. grep '^[c]ode' p
 4. grep -E "([a-zA-Z]+)" stud
 5. sed -E 's/\$/\$\$/p' stud
 6. sed -n '3,7p' p|sed '/^ *\$/d'
 7. grep -l 'bc' p|wc -l
-

1. Display the line which begin and end with same character of file x1.
grep '^(\.)*\1\$' x
2. Write the command to display lines which do not contain word surat.
grep -v 'Surat' x
3. Change all the words "Baroda" with "Vadodara".

Sed -n 's/Baroda/Vadodara/g' x

4. Insert ":" After every line

sed 's/\$/./' x

5. Write a command to print line with line number which contain "BCA".

grep -n 'bca' x

6. Display the occurrences of each word.

tr -s " " "\012" <b|sort|grep -c

7. Display the line whose length is exact 50.

sed -ne '/^.\{50\}\$/p' x

8. Display the word whose length is between 5 to 10 characters.

sed -ne '/^.\{5,10\}\$/p' x

9. Display lines which end with ".".

grep '\.\$' x

10. Write the command to display first three line of file.

sed -n '1,3p' x

11. Display the lines which do not begin with vowel.

grep -e '^[^aeiou].*' x

12. Copy the lines in x2 from x1 which has only 2 digits.

grep '^'[0-9][0-9] '\$' x > x1

13. Display all regular files of working directory whose permission for all user are read and write.

ls -l | sed -n '/^-\{1,3\}rw./p'

ls -l | sed -n '/^-\(rw.\)\1\1/'

14. Display all file of working directory whose file name consists of only alphabet.

ls | grep '^'[a-zA-Z.] '\$'

15. Count the number of words in line 40-60 of file f1.

sed -n '40,60p' b | tr -s " " "\012" | wc -l

16. To list the file name consist of only 4 digits.

ls | grep '[0-9]\{4\}'

17. Replay 'kernel' with 'kernel architecture' using remember pattern of sed utility.

```
Sed '/kernel/s//& architecture/' x
```

18. Display the line having at least one '*' character in file x1

```
grep '[*]\\{1,\\}' x
```

Or

```
Grep '\\*\\{1,\\}' x
```

19. Display line from file x1 , that contains "UNIX", "unix" or 'Unix'.

```
Cat b1
```

```
UNIX
```

```
Unix
```

```
unix
```

```
grep -n -f b1 b
```

or

```
sed -n '/unix/I p' b1
```

20. Write command to display occurrences of string "Sales"

```
tr -s "|" "\\012" <b |sort|uniq -c|grep 'sales'
```

21. Display 5 letters palindrome words from file x1.

```
grep '^\\.\\.\\.\\.\\.\\2\\1$' x
```

22. Display the lines which contain two or more \$ sign at end of line.

```
grep '\\${2,}\\$' b
```

23. Print the line which contains consecutive "Hello" two times in a line.

```
grep '\\(hello\\) \\1' b
```

24. Display name of files of working directory having same permission to owner, group and other user.

```
ls -l | sed -n '/^\\.\\.\\.\\1\\1/p'
```

25. To count one character words present in file f1.

```
grep '^\\[a-zA-Z0-9]\\{1\\}$' b
```

26. Display those words in which any characters repeated constantly at least 2 times.

```
Grep '\\(.\\*\\) \\1' stud
```

27. List of empty files.

```
ls -l | tr -s ' ' | grep '^(\.*) \(\.*) \(\.*) \(\.*) 0 \.*$'
```

28. Add two spaces to beginning of each line and 2 dashes to end of each line

```
Sed -n "s/^/  /" b1 | sed -n "s/$/--/"
```

29. To remove all trailing spaces from a file.

```
sed -E 's/\ +$//g' yy1
```

30. Change '/' to '-' in date and convert it to mm-dd-yyyy format.

```
sed 's/\(..\)\/\(..\)\/\(\...\)\/\2-\1-\3/' x
```

31. Display the lines in which 2nd and 4th words are same.

```
tr -s ' ' <xyz | grep '^(\.*) \(\.*) \(\.*) \2.*$'
```

1. Write a ^{menu} driven shell script to perform 30 following operation

1. Copy file 2. Remove 3. Move 4. quit

```
echo "Menu "
```

```
echo "1. COpy a File "
```

```
echo "2. Remove a file "
```

```
echo "3. Move a file"
```

```
echo "4. Quit"
```

```
echo "Enter ur Choice \c"
```

```
read Choice
```

```
case $Choice in
```

```
1) echo "Enter File name to copy \c"
```

```
read f1
```

```
echo "Enter File name \c "
```

```
read f2
```

```
if [ -f $f1 ]
```

```
then
```

```
cp $f1 $f2
```

```
else
```

```
    echo "$f1 does not exist"
```

```
fi
```

```
;;
```

```
2) echo "Enter the File to be removed "
```

```
    read r1
```

```
    if [ -f $r1 ]
```

```
    then
```

```
        rm -i $r1
```

```
    else
```

```
        echo "$r1 file does not exist "
```

```
    fi
```

```
;;
```

```
3)
```

```
    echo "Enter File name to move \c"
```

```
    read f1
```

```
    echo "Enter destination \c "
```

```
    read f2
```

```
    if [ -f $f1 ]
```

```
    then
```

```
        if [ -d $f2 ]
```

```
        then
```

```
            mv $f1 $f2
```

```
        fi
```

```
    else
```

```
        echo "$f1 does not exist"
```

```
    fi
```

```
;;
```

```
4)
```

```
    echo "Exit....."

    exit;;

esac
```

2. Decimal to Binary 2. Decimal to Octal 3. Decimal to Hexadecimal

```
echo "Menu "

echo "1. decimal to binary "
echo "2. decimal to oct "
echo "3. decimal to hexa"
echo "4. binary to decimal"
    echo "5. octal to hexa"
    echo "6. hexa to decimal"

echo "7. Quit"

echo "Enter ur Choice \c"

read Choice

case $Choice in

    1) echo "decimal to binary \c"

        echo " enter the decimal number"

        read b

        echo "ibase=10;obase=2; $b" | bc

        ;;

    2) echo "decimal to oct \c"

        echo " enter the decimal number"

        read b

        echo "obase=8; $b" | bc

        ;;

    3)

        echo "decimal to hexa \c"
```



```

        echo " enter the decimal number"

        read b

        echo "ibase=10;obase=16; $b" | bc

        ;;

4) echo "binary to decimal \c"

        echo " enter the decimalnumber"

        read b

        echo "ibase=2;obase=10; $b" | bc

        ;;

5) echo "octal to decimal\c"

        echo " enter the oct number"

        read b

        echo "ibase=8; $b" | bc

        ;;

6) echo "hexa to decimal\c"

        echo " enter the hexa number"

        read b

        echo "ibase=16; $b" | bc

        ;;

7) echo "Exit....."

        exit;;

    esac

```

1. Display 3 to 10 lines from file x 2. Display total no of blank line 3.remove special characters from file x1 4. Quit

```

echo "Menu "

echo "1. Display line from 3-10 "

echo "2. Display the number of blank lines "

echo "3. remove spacial character"

echo "4. Quit"

```

```

echo "Enter ur Choice \c"

read Choice

case $Choice in

    1) echo "display 3 to 10 \c"

        sed -n '3,10p' x

        ;;

    2) echo "display no of blank line\c "

        grep "^ *$" x | wc -l

        ;;

    3)

        echo "remove special character \c"

        tr -cd '[a-z][A-Z][0-9]'<x

        ;;

    4)

        echo "Exit....."

        exit;;

esac

```

Write shell program to check user given character is in small letter, capital letter, digit or special character.

```

Echo "enter character"

Read ch

case $ch in

[a-z]) echo "it is small letter"

;;

[A-Z]) echo "it is capital letter"

;;

[0-9]) echo "it is digit"

;;

?) echo "it is special letter"

```

<pre>;; *) echo "it is more than one letter" ;; Esac</pre>
Write shell program to check user given string start with small case vowel, capital case vowel, end with digit or three letters word.
<pre>Echo "enter character" Read ch case \$ch in [aeiou]*) echo "begins with small case vowel" ;; [AEIOU]) echo "begins with small case vowel" ;; *[0-9]) echo "End with digits" ;; ???) echo "three letter word" ;; *) echo "invalid input" ;; Esac</pre>

Write a shell script that will pass filename and two 10 integer numbers n1 and n2. Display lines n1 to n2 from file
<pre>S1.sh Sed -ne "\$2,\$3p" \$1 -run sh s1.sh stud 3 5</pre>
Write a UNIX shell script that copy a file to another file and reads file content in-reverse. Try to incorporate as many files is possible.
<pre>a=`cat -n x tail -n 1 cut -f 1`</pre>

```
while [ $a -ge 1 ]  
  
do  
  
cat x | head -n $a | tail -n 1 >> l  
  
let a=a-1  
  
done
```

Write Unix shell script that display content of file line by line. It displays total words and characters in each line of the file. At the end, it should display, total number of characters in the file and total lines.

```
echo Enter the filename  
  
read file  
  
a=`cat -n $file | tail -n 1 | cut -f 1`  
  
while test $a -ge 1  
  
do  
  
cat $file | head -n $a | tail -n 1  
  
w1=`cat $file | head -n $a | tail -n 1 | wc -w`  
  
c1=`cat $file | head -n $a | tail -n 1 | wc -c`  
  
echo "No of words in $a line are $w1 and characters are $c1"  
  
let a=a-1  
  
done  
  
w=`cat $file | wc -w`  
  
c=`cat $file | wc -c`  
  
l=`grep -c "." $file`  
  
echo Number of characters in $file is $c  
  
echo Number of words in $file is $w  
  
echo Number of lines in $file is $l
```

Shell script for Calculate +,-,* or / of two numbers[using command line argument]

```
if test $# = 3  
  
then  
  
    case $2 in
```

```

+) z=`expr $1 + $3`;
-) z=`expr $1 - $3`;
/) z=`expr $1 / $3`;
x|X) z=`expr $1 \* $3`;
*) echo Warning - $2 invalied operator, only +,-,x,/ operator allowed
exit;;
esac
echo Answer is $z
else
echo "Usage - $0 value1 operator value2"
echo "    Where, value1 and value2 are numeric values"
echo "    operator can be +,-,/,x (For Multiplication)"
fi

```

Check given user name is log in or not [using command line argument]

```

echo "enter user name"
read user
who | grep $user >/dev/null
if test $? -eq 0
then
echo "it is exists"
else
echo "it is not exists"
fi

```

Shell script for print 1 to 10, odd number and even numbers[between 1 to 10]

```

# 1 to 10
for i in `seq 1 10`
do
echo $i

```

```
done

# odd numbers
echo "odd"
for j in `seq 1 2 10`
do
    echo $j
done
```

```
# even numbers
echo "even"
for j in `seq 2 2 10`
do
    echo $j
done
```

Write shell script for check palindrome or not

```
Clear
echo "Enter a string to be entered:"
read str
echo
len=`echo $str | wc -c`
let len=len-1
i=1
let j=len/2

while [ $i -le $j ]
do
    k=`echo $str | cut -c $i`
```

```
l=`echo $str | cut -c $len`
```

```
if [ $k != $l ]
```

```
then
```

```
echo "String is not palindrome"
```

```
exit
```

```
fi
```

```
let i=i+1
```

```
let len=len-1
```

```
done
```

```
echo "String is palindrome"
```

Write command line shell script for reverses the number.

Filename ->> rev.sh

```
if [ $# -ne 1 ]
```

```
then
```

```
    echo "Usage: $0 number"
```

```
    echo "    I will find reverse of given number"
```

```
    echo "    For eg. $0 123, I will print 321"
```

```
    exit 1
```

```
fi
```

```
n=$1
```

```
rev=0
```

```
sd=0
```

```
while [ $n -gt 0 ]
```

```
do
```

```
    sd=`expr $n % 10`
```

```
    rev=`expr $rev \* 10 + $sd`
```

```
    n=`expr $n / 10`
```

```
done
    echo "Reverse number is $rev"
```

run-> sh rev.sh 1234

or

```
if [ $# -eq 1 ]
then
    if [ -f $1 ]
    then
        a=`rev $1`
        echo "Reverse of $1"
        cat $1
        echo " is-> $a"else
        echo "File does not exist "
    fi

else
    echo "Please enter file name or path"
fi
```

run-> sh rev1.sh 1234

Write shell script for searching given pattern from given file

```
echo "pattern to be search:\c"
read pname
echo "enter fname\c"
read fname
grep "$pname" $fname
```

Write shell script for searching given pattern from given file [using command line argument]

```
echo "program : $0"
```



```
echo "argument:$#"
echo " no of argument : $*"
grep "$1" $2
echo "finished"
run sh s2.sh Surat x
```

Write shell script for Number of vowels in the string

```
echo "Enter a string:"
read str
count=$(echo "$str" | grep -o -i "[aeiou]" | wc -l)
echo "Number of vowels in the string: $count"
```

Write shell script for Toggle case

```
echo "Enter a string:"
read str
toggle=$(echo "$str" | tr 'a-zA-Z' 'A-Za-z')
echo "Toggled case: $toggle"
```

a **shell script** that counts:

- Lowercase letters
- Uppercase letters
- Digits
- Special characters

```
echo "Enter a string:"
read str
lower=$(echo "$str" | grep -o "[a-z]" | wc -l)
upper=$(echo "$str" | grep -o "[A-Z]" | wc -l)
digit=$(echo "$str" | grep -o "[0-9]" | wc -l)
special=$(echo "$str" | grep -o "[^a-zA-Z0-9]" | wc -l)
echo "Lowercase letters : $lower"
echo "Uppercase letters : $upper"
```

```
echo "Digits      : $digit"
echo "Special characters: $special"
```

a shell script to display even numbers between two numbers.

```
echo "Enter the starting number:"
read start
echo "Enter the ending number:"
read end
echo "Even numbers between $start and $end are:"
for (( i=$start; i<=$end; i++ ))
do
    if [  $$(i \% 2)$  -eq 0 ]; then
        echo -n "$i "
    fi
done
echo
```